Fiber Optics

James T. Aberle, Ph.D.
Associate Professor Emeritus

School of ECEE

Ira A. Fulton Schools of Engineering
Arizona State University

Lecture Objective

By the end of this lecture, you will be prepared to:

- Evaluate metrics like numerical aperture and bandwidth for an optical fiber.

Fiber Optic Waveguide

- Electromagnetic waves can be guided through dielectric rods by successive total internal reflections.
- Such a structure is an optical fiber, a form of dielectric waveguide.

2. Fiber Optic Waveguide

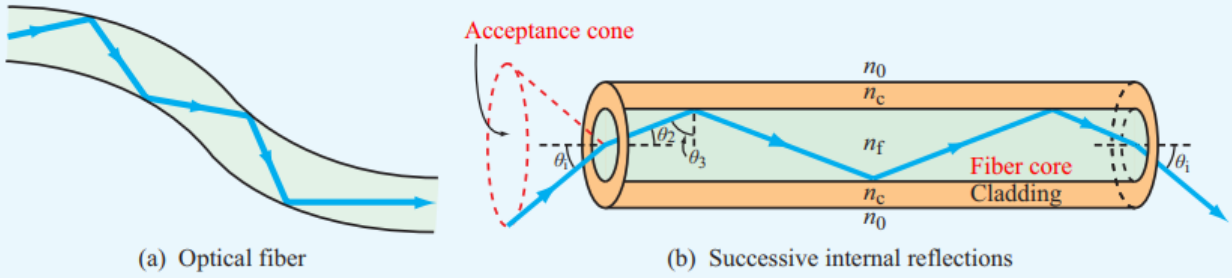


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

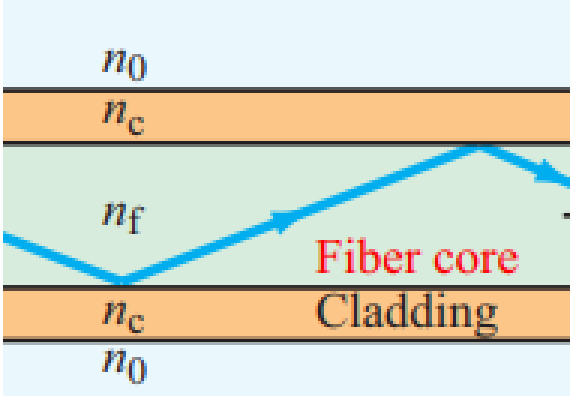
3. Fiber Optic Waveguide

- Recall that when total internal reflection occurs at a boundary, the electromagnetic field on the other side of the boundary is evanescent (not zero).
- Therefore, it is possible for electromagnetic energy to couple between adjacent fibers.

Fiber Core and Cladding

- Thus, an optical fiber usually comprising a dielectric core surrounded by a dielectric cladding.
- The cladding is designed to isolate the optical fiber from others in close proximity.
- For total internal reflection,

$$n_f > n_c$$



$$n_0 = 1$$

Condition for Total Internal Reflection

- For total internal reflection, we must have

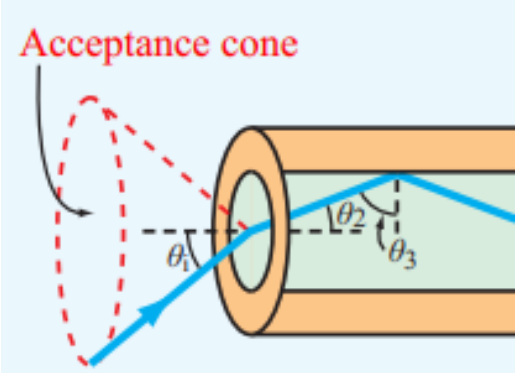
$$\theta_3 > \theta_c = \operatorname{asin}\left(\frac{n_c}{n_f}\right)$$

- Since $\theta_2 + \theta_3 = \dfrac{\pi}{2}$

$$\cos\theta_2 = \sin\theta_3$$

- The necessary condition for total internal reflection becomes

$$\cos\theta_2 > \frac{n_c}{n_f}$$



Acceptance Cone

- Furthermore, the incidence angle on the fiber face (θ_i) is related to θ_2 by Snell's law as

$$\sin\theta_2 = \frac{n_0}{n_f}\sin\theta_i \quad \Rightarrow \quad \cos\theta_2 = \sqrt{1-\left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i}$$

$$\sqrt{1-\left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i} \geq \frac{n_c}{n_f} \quad \Rightarrow \quad \sin\theta_i \leq \frac{1}{n_0}\sqrt{n_f^{\,2} - n_c^{\,2}}$$

Acceptance Cone and Numerical Aperture

- Any ray of light incident upon the fiber face within the acceptance cone can propagate down the fiber core.
- The cone angle of the acceptance cone is $2\theta_a$ where θ_a is the maximum value of θ_i for which total internal reflection occurs.
- The numerical aperture ($N.A.$) is the sine of the acceptance angle.

$$N.A. = \sin\theta_a = \frac{1}{n_0}\sqrt{n_f^{\,2} - n_c^{\,2}}$$

Modes

- Light rays can propagate along the optical fiber at various angles.
- The velocity of propagation for any given ray is

$$u_p = c/n_f$$

- The axial mode propagates along the axis of the fiber without reflections.
- Hence, the axial mode is the one that travels the shortest path through the cable of length l and its travel time is the shortest possible given by

$$t_{\min} = \frac{l}{u_p} = \frac{l}{c} n_f$$

2. Modes

- The mode that reflects from the boundary at the critical angle travels the longest path through the cable of length l and its travel time is the maximum possible given by

$$t_{\max} = \frac{l/\sin\theta_c}{u_p} = \frac{l}{c}\frac{n_f}{\sin\theta_c} = \frac{l}{c}\frac{n_f^{\,2}}{n_c}$$

- The total time spread of the response to an impulse is given by

$$\tau = t_{\max} - t_{\min} = \frac{l}{c}\frac{n_f^{\,2}}{n_c} - \frac{l}{c}n_f = \frac{l}{c}n_f\left(\frac{n_f - n_c}{n_c}\right)$$

3. Modes

- An impulse in time has a flat frequency response from $\omega = 0$ to $\omega \rightarrow \infty$.
- The frequency at which the ray propagates as the critical angle is the lowest frequency which can propagate in the fiber (“cutoff frequency”).
- As frequency gets higher and higher, the ray propagation tends toward the axial mode.
- The propagation of different frequency components at different velocities results in dispersion.

Intermodal Dispersion

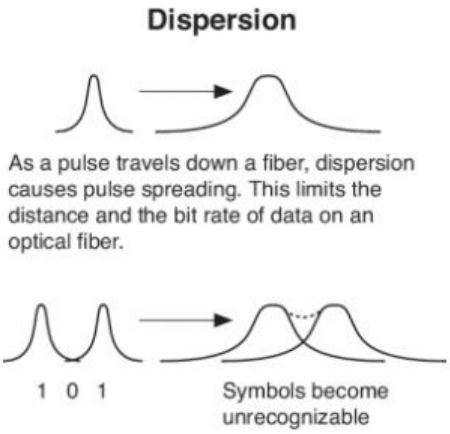


Figure 4. Dispersion in an optical fiber.

https://www.newport.com/t/fiber-optic-basics

2. Intermodal Dispersion

- Dispersion results in time spread of the impulse response.
- Dispersion happens in a waveguide whenever different frequency components of an information bearing signal propagate at different velocities.
- Dispersion is the mechanism that limits the bandwidth of the waveguide.

3. Intermodal Dispersion

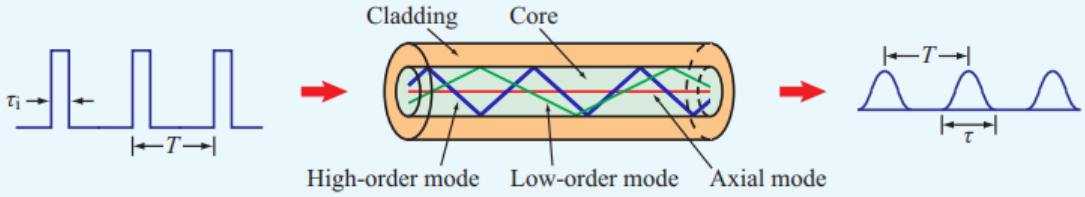


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

- As a rule-of-thumb, the system is designed so that the interpulse period (T) is greater than twice the delay spread (τ).

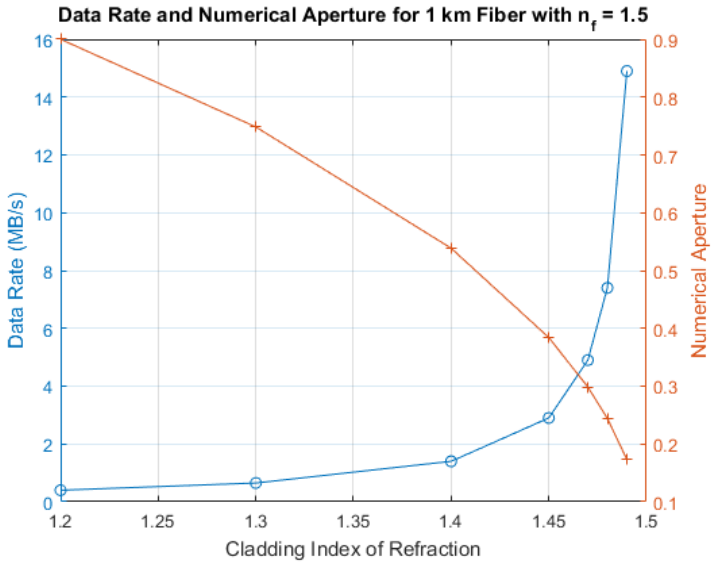
Bandwidth

- Thus, the data rate (aka “bandwidth”) of the system can be no greater than

$$f_p = \frac{1}{T} \leq \frac{1}{2\tau} = \frac{c}{2l}\left(\frac{n_c/n_f}{n_f - n_c}\right)$$

- Other mechanisms that limit bandwidth include chromatic dispersion resulting from a change in index of refraction (and hence, phase velocity) versus frequency.
- Factors that can increase bandwidth include
  - Shorter cable length
  - Cladding index of refraction closer to core index of refraction

Bandwidth and Numerical Aperture Versus Cladding Index of Refraction



Question

The performance of specific optical fiber products vis-à-vis intermodal dispersion are often quantified using a figure-of-merit called bandwidth-length (BL) product. Which reason below explains why this figure-of-merit is especially useful?

(a) This FoM depends only on the material properties and not the length of the cable.

(b) This FoM accounts for chromatic dispersion.

(c) This FoM is independent of the index of refraction of the cladding.

(d) This FoM predicts the probability of detection for specific data packets.

Comments on Numerical Aperture

- Optical fibers with large N.A. are less sensitive to alignment tolerances and are used in applications where high efficiency is tantamount.
- Optical fiber with small N.A. are used in applications for which high data rates and/or long interconnects are needed.

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